Economic Opportunity & Urban Development Analysis of the Chicago area

Bhavisha Ahuja, Dima Medvedkov, Filip Petrev,
Muhammad Taha, Sai Rohit Mahadas

Motivation

Geography of Economic Outcomes

Map where poverty and unemployment cluster, then examine which community features co-locate with those outcomes.

Which Factors Matter Most?

Are structural factors—access to quality schools, transit, and local businesses—more predictive of unemployment and poverty than crime?

Pinpointing the Opportunity Divide

Chicago's economic outcomes are uneven across neighborhoods; we aim to measure those gaps.

Data

Source: U.S. Census Bureau's American Community Survey (ACS)

What is the ACS?

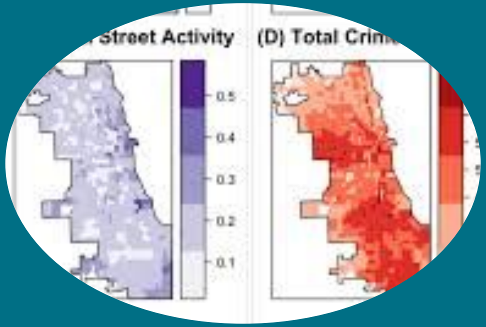
- The ACS is the Census Bureau's continuous, nationwide sample survey (monthly collection, annual release)
- It provides estimates with 90% confidence intervals for social and economic indicators.

What will we collect?

- Unemployment rate (%)
- Poverty rate (% below poverty)
- Educational attainment (high school, bachelor's+)
- Median household income ($)
- Labor force participation rate (%)
- Housing cost burden (%)
- Population density

AMERICAN COMMUNITY SURVEY
U.S. CENSUS BUREAU

Solution #1 - Geospatial Visualization



Interactive Choropleth Maps

We will create a series of interactive choropleth maps of Chicago that visualize the geographic distribution of key socio-economic variables at the census tract level.

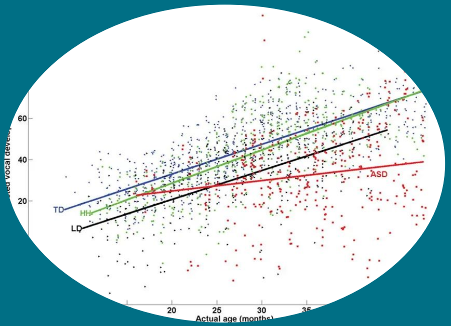
Key Variables:

- Unemployment Rate
- Poverty Level
- Public Transit Stop Density
- Local Business Density

Users will be able to select different variables to display on the map, zoom into specific neighborhoods, and hover over a census tract to see its specific data points.

This will provide a clear, visual understanding of where economic opportunity disparities exist in the city and how they spatially correlate with community resources.

Solution #2 - Predictive Modeling



Multiple Linear Regression

We will build a statistical model that quantifies the relationships between neighborhood characteristics and economic outcomes.

Dependent Variable (The outcome we want to predict):

- Unemployment Rate per census tract.

Independent Variables (The factors we think are predictive):

- Transit access (e.g., distance to nearest 'L' station).
- Density of businesses.
- Proximity to schools.

The model will identify which factors are the most statistically significant predictors of unemployment, helping us move beyond simple correlation to understand the underlying drivers of economic disparity.

Goals

1. Create an interactive visualization, allowing users to explore findings and make decisions based on them on the local scale.

2. Demonstrate result of developed model, showcasing how different features predict economic hardship.